

ECP 7408 Empirical Industrial Organization I

Semester:	Fall 2025	Office Hours:	By Appointment
Instructor:	Douglas Turner	Office:	MAT 337
Email:	douglasturner@ufl.edu	Class Hours:	MW 1:55-3:50
Classroom:	MAT 0018 (Mondays) MAT 0105 (Wednesdays)	TA:	Qilin Zhang qilin.zhang@ufl.edu

Course Description

ECO 7408 provides a Ph.D. level introduction to empirical industrial organization with a focus on demand estimation. Students will explore the tools and techniques used to analyze consumer choice and firm behavior in differentiated product markets. The course covers discrete choice models, including logit and nested logit, and advances to more flexible approaches such as random coefficients models (e.g., BLP). Dynamic demand estimation is also covered. Emphasis is placed on the identification and estimation of demand systems, the use of instrumental variables, and applications. Students will engage with both the theory, econometrics and practical implementation of these models using data.

Course Materials

There is no required textbook for this course. However, the following textbooks may provide helpful references:

Train, Kenneth E. Discrete Choice Methods with Simulation. Second Edition. 2009. Cambridge University Press.

Aguirregabiria, Victor. Empirical Industrial Organization: Models, Methods, and Applications.

http://aguirregabiria.net/wpapers/book_dynamic_io.pdf

Exams

There will be one exam, on paper and in class, which is tentatively scheduled for 12/2. The exam is cumulative. Exams are closed book and closed notes.

Problem Sets

- There will be problem sets assigned periodically throughout the semester. These assignments are weighted equally. No late assignments are permitted.
- Solutions to these assignments should be written in LaTeX and submitted as a PDF. Code should be included with your submission. LaTeX is a software system for professionally typesetting documents and mathematical equations. If you have not used LaTeX before, I recommend beginning with LyX. LyX is a free, easy to use application for creating documents using LaTeX. See <https://www.lyx.org> for more information on LyX and <https://www.lyx.org/Download> to download LyX.
- Some problem sets will require the use of a programming language. This course will use the python programming language. Python is a widely used programming language known for its readability and versatility in data analysis, web development, automation, and more. To set it up on your computer, install the latest version from python.org, and consider using Anaconda for an all-in-one package that includes Python, Jupyter Notebook, and many useful data science libraries. Once installed, you can run Python code through a terminal, text editor, or interactive environment like Jupyter.
- You are encouraged to work with other students on the problem sets, but each student must write up his or her answers separately.

Project and Presentation

- The course will involve a project where you will find data suitable for demand estimation, estimate a demand model and analyze your results. The project will result in 1) a research paper presenting and analyzing your results and 2) a presentation to the class. More details on the project will be given as the semester proceeds. Each student is responsible for submitting their own project.

Participation

- Students are expected to attend class and engage with the lecture. Engaging with the lecture involves asking questions and also attempting to answer questions posed by the instructor. You are not evaluated on the quality of your questions or answers, only on your willingness to attempt to ask/answer questions.

Grading

- Any curving will be at the discretion of the instructor. The grading scheme is

Grading Scheme		Final Grading Scale			
		A	92-100	C	72-76.99
Assignments	20%	A-	90-91.99	C-	70-71.99
Participation	5%	B+	87-89.99	D+	67-69.99
Final Exam	35%	B	82-86.99	D	60-66.99
Project	40%	B-	80-81.99	E	0-59.99
		C+	77-79.99		

Course Schedule

MONDAY	WEDNESDAY
8/25 Demand Models Intro Linear Demand Models	8/27 Linear Demand Models Logit
9/1 NO CLASS (Labor Day)	9/3 Logit Multinomial Logit
9/8 Multinomial Logit (Aggregate Market Share Version)	9/10 Multinomial Logit (Aggregate Market Share Version)
9/15 Nested Logit	9/17 Demand Elasticities and Consumer Welfare
9/22 Python Lab	9/24 Random Coefficients Logit
9/29 Random Coefficients Logit	10/1 NO CLASS (Project Work Day)
10/6 Random Coefficients Logit	10/8 Random Coefficients Logit

MONDAY	WEDNESDAY
10/13 Python Lab	10/15 Random Coefficients Logit
10/20 Random Coefficients Logit	10/22 Dynamic Demand
10/27 Dynamic Demand	10/29 Dynamic Demand
11/3 Dynamic Demand	11/5 Dynamic Demand Presentations
11/10 No Class (Veteran's Day)	11/12 Presentations
11/17 Presentations	11/19 Presentations
11/24 NO CLASS (Holiday)	11/26 NO CLASS (Holiday)
12/1 Exam Review	12/3 Exam

Academic Honesty

Students are expected to comply with the University of Florida's rules for academic honesty. Failure to comply with this commitment will result in disciplinary action.

Students are bound to not cheat or plagiarize, and are bound by The Honor Pledge which states, "We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: 'On my honor, I have neither given nor received unauthorized aid in doing this assignment.'"

You should familiarize yourself with the UF Student Honor Code. Cheating and plagiarism are not the only violations of this policy. Importantly, ignorance of a policy is not a valid reason for violating it.

Accommodations for Students with Disabilities

Students with disabilities requesting classroom accommodation must first register with the Disability Resource Center. The Disability Resource Center will provide documentation to the student who must then provide this documentation to the instructor when requesting accommodation. Students with disabilities should follow this procedure as early as possible in the semester.

Online Course Evaluations

Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Guidance on how to give feedback in a professional and respectful manner is available at <https://gatorevals.ua.ufl.edu/students/>. Students will be notified when the evaluation period opens, and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via <https://ufl.bluer.com/ufl/>. Summaries of course evaluation results are available to students at <https://gatorevals.ua.ufl.edu/public-results/>.

UF Teaching Center

The UF Teaching Center offers guidance on study skills and tutoring services. You can find more information at: <https://umatter.ufl.edu/office/teaching-center/>.

Generative Artificial Intelligence

The Department of Economics faculty assume that all work that is submitted for grading is written by the student whose name it bears, and that it represents their ideas and work. Accordingly, students are not permitted to use generative AI when completing assignments, quizzes, exams, or other graded work unless their instructor has expressly granted that permission. Unauthorized use of generative AI may constitute cheating and/or plagiarism. Such violations of the UF Student Honor Code will be reported to the UF Dean of Students Office and will be subject to severe sanctions.

Class Attendance and Make-Up Exams

Requirements for class attendance and make-up exams, assignments, and other work in this course are consistent with university policies. Click here to read the university attendance policies.

Make-up assignments and exams will be arranged only for absences that are explicitly covered by the UF Attendance Policy. Whenever possible, you should reach out at least five business days in advance to arrange a make-up assignment or exam. Of course, this will not always be possible. Unforeseen absences and emergencies occur and can be excused without such advance notice. In most cases, you will be asked to provide evidence or documentation of an absence that is explicitly excused by the UF Attendance Policy. Absences related to religious holidays and worship do not require this documentation.

Resources

- **U Matter, We Care:** If you or a friend is in distress, please contact umatter@ufl.edu or 352 392-1575 so that a team member can reach out to the student.
- **Counseling and Wellness Center:** <http://www.counseling.ufl.edu/cwc>, and 392-1575; and the University Police Department: 392-1111 or 9-1-1 for emergencies.
- **Sexual Assault Recovery Services (SARS):** Student Health Care Center, 392-1161.
- **University Police Department** at 392-1111 (or 9-1-1 for emergencies), or <http://www.police.ufl.edu/>.

References

Introduction

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Static Demand Models with Homogeneous Goods

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